AR-Mini · Where each job lives

Claude thinks and plans. Cursor writes the files. ChatGPT and Gemini check. Teddy builds and measures. iPhone and the tracker are the proof.

Claude

Claude Max
think · plan

- Coordinator: plan for the day
- Mechanical + Electrical: decisions and hypothesis
- Test & Data: CSV → pass / fail
- Logistic: BOM, orders, deliveries
- Portfolio: texts, FAILS_AND_FIXES
- posters, visual BOM, diagrams

example prompt

"Here is today's tracker. What are my 3 priorities?"

→ plan · hypothesis · checklist

CURSOR

Cursor Premium
write files · commit

- AR-Mini_v1.scad parameters
- interfaces.json → sync to CAD
- KiCad files + ERC / DRC
- firmware in PlatformIO
- scripts: CSV → charts
- commit + push to dev

example prompt

"Arm 110 → 115 mm. Follow the rules, run check, commit to dev."

→ files in the repo

ChatGPT

ChatGPT Plus
second opinion

- before the first 12 V power
- before a long PETG print
- photo of the wiring: what is wrong?
- voice mode while soldering

example prompt

[photo] "I am about to power this with 12 V. What can burn?"

→ a list of risks

Gemini

Gemini Plus
read · watch

- long datasheets (ESP32-S3, BTS7960, MLX90640)
- test video, frame by frame
- time stamp of leak or tilt
- short summary → Claude

example prompt

[video] "At which second does the hull start to tilt?"

→ facts with a source

By hand

By hand
you are faster here

- Fusion 360: FEA, renders, STEP fit check
- KiCad: routing, 3D view
- Creality Print: slicing
- OpenSCAD: F5 preview
- printing, assembly, soldering, measuring

tip

Route power traces first, then signals. Check the STEP in Fusion before ordering.

→ parts on the bench

iPhone

iPhone + tracker
the proof

- photo: part next to a ruler
- photo: meter display for every number
- 4K 30 fps video, slo-mo for water
- AR-Mini Loop: log every loop

shot

Part + ruler, light from the side, phone flat above. One clip per loop.

→ numbers for the portfolio

GitHub

GitHub
history

- dev ← Cursor commits
- main ← Teddy, after the test
- Actions: OpenSCAD, ERC / DRC, firmware build
- Gerber, drill, STEP exported

rule

Open the PR dev → main only after the test is logged in the tracker.

→ real dates, every version

One loop, step by step

Red = Teddy. Every arrow passes through her.

1 Plan — Coordinator: what today? — Claude

→ **2 Decide** — Mechanical / Electrical: change + hypothesis — Claude

→ **3 Check** — risky step? ask for the risks — ChatGPT

→ **4 Write** — "make the change, check, commit to dev" — CURSOR

→ **5 Build + measure** — print, assemble, wire, test, film — iPhone

→ **6 Analyse** — video + CSV → pass / fail — Gemini · Claude

→ **7 Close** — log in the tracker; pass → merge dev → main — tracker + GitHub

next version

One home per job

Only Cursor writes repo files. Claude plans. Two tools on one file will fight. If Claude has an idea for a file, it goes to Cursor as a task.

Carry a summary, not the chat

ChatGPT and Gemini do not know the project: give 5 lines of context + the question. Bring the answer back to Claude in 2–3 lines.

Hands stay on three things

KiCad routing, Fusion FEA and slicing are faster and safer by hand. AI can review the result from a screenshot.